Day 7: Training Flow & Diffusion Models

Welcome

- **Training Flow & Diffusion Models** — Continuous transforms and denoising objectives
- 3 hours lecture + practical
- Slides, notes, and code on the course site

Outline

- Normalizing Flows
- Diffusion Intuition
- Score Matching View
- Training Practice

Normalizing Flows

Change of Variables

- $p_{X(x)} = p_{Z(f^{-1}(x))} \; |\det J_{f^{-1}}(x)|$
- Bijective f with tractable inverse
- Composition of simple coupling layers

Lecture 2

Flow Matching

MIT IAP 2026 | Jan 22, 2025

Peter Holderrieth and Ron Shprints

Sponsor: Tommi Jaakkola

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Normalizing Flows — Change of Variables (source: course materials)

Coupling Layers

- Split dimensions; transform one part conditioned on other
- Affine coupling: scale and shift
- RealNVP, Glow architectures

Next step: Training a flow model

Without training, the model produces "non-sense"→ **We need to train** u_t^θ

Training = Finding parameters θ **such that**

$X_0 \sim p_{\mathrm{init}}, \quad \mathrm{d}X_t = u_t^\theta(X_t)\mathrm{d}t$ Implies $X_1 \sim p_{\mathrm{data}}$

Start with initial distribution

Follow along the vector field

The distribution of the final point = data distribution

Goal of lecture 2 (today):

Derive Flow Matching (training algorithm for flow models)

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Normalizing Flows — Coupling Layers (source: course materials)

Training Flows

- Maximize log-likelihood directly
- Jacobian log-determinant cost
- Exact sampling and density

The Flow Matching Matrix

Conditional Probability Path →	**Conditional** Vector Field →	**Conditional** Flow Matching Loss
Marginal Probability Path →	**Marginal** Vector Field →	**Marginal** Flow Matching Loss

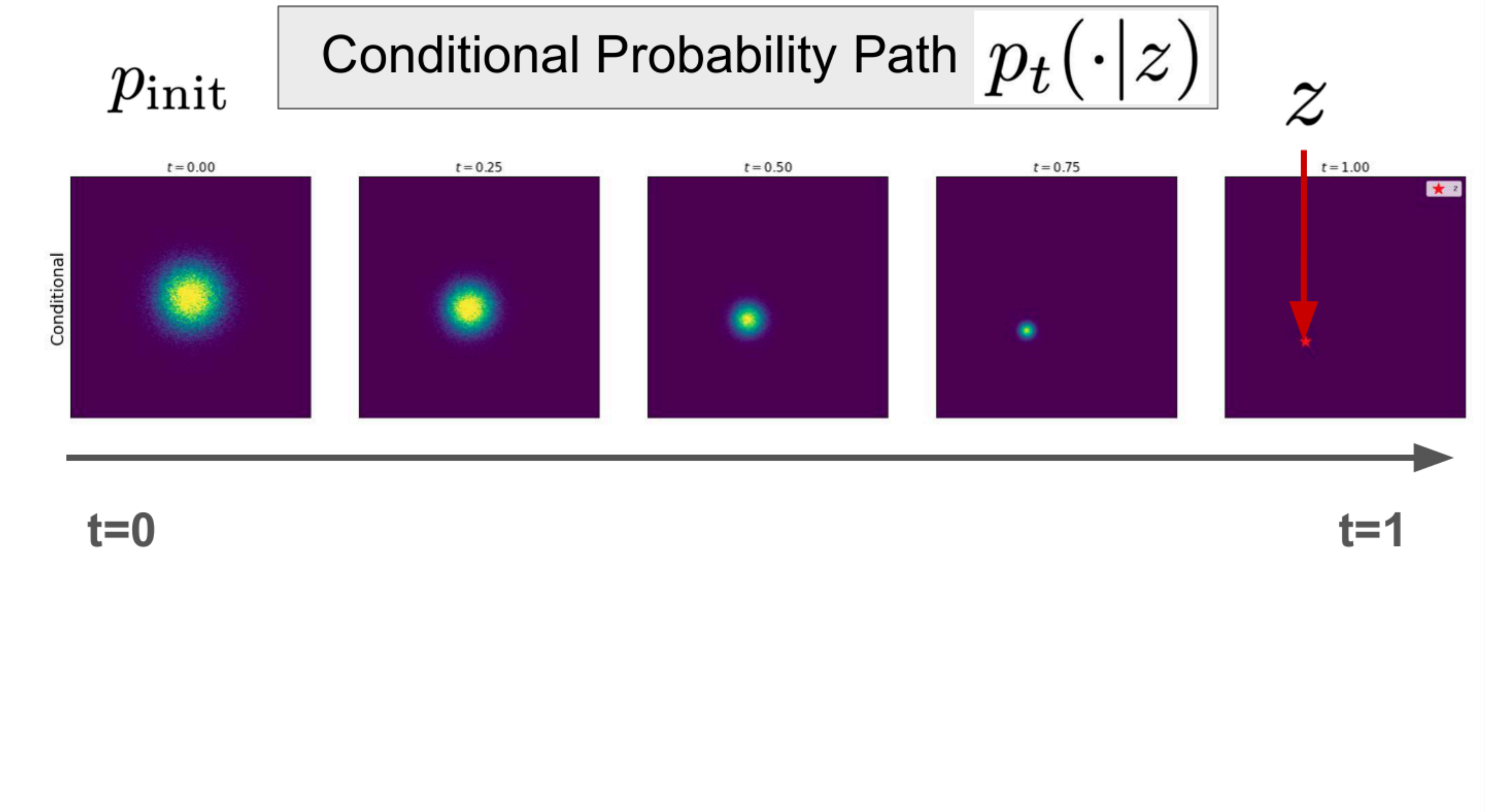
“Conditional” = “Per single data point”

“Marginal” = “Across distribution of data points”

Normalizing Flows — Training Flows (source: course materials)

Limitations

- Architectural constraints for invertibility
- Scaling to high-res images is hard
- Diffusion trades exact likelihood for flexibility

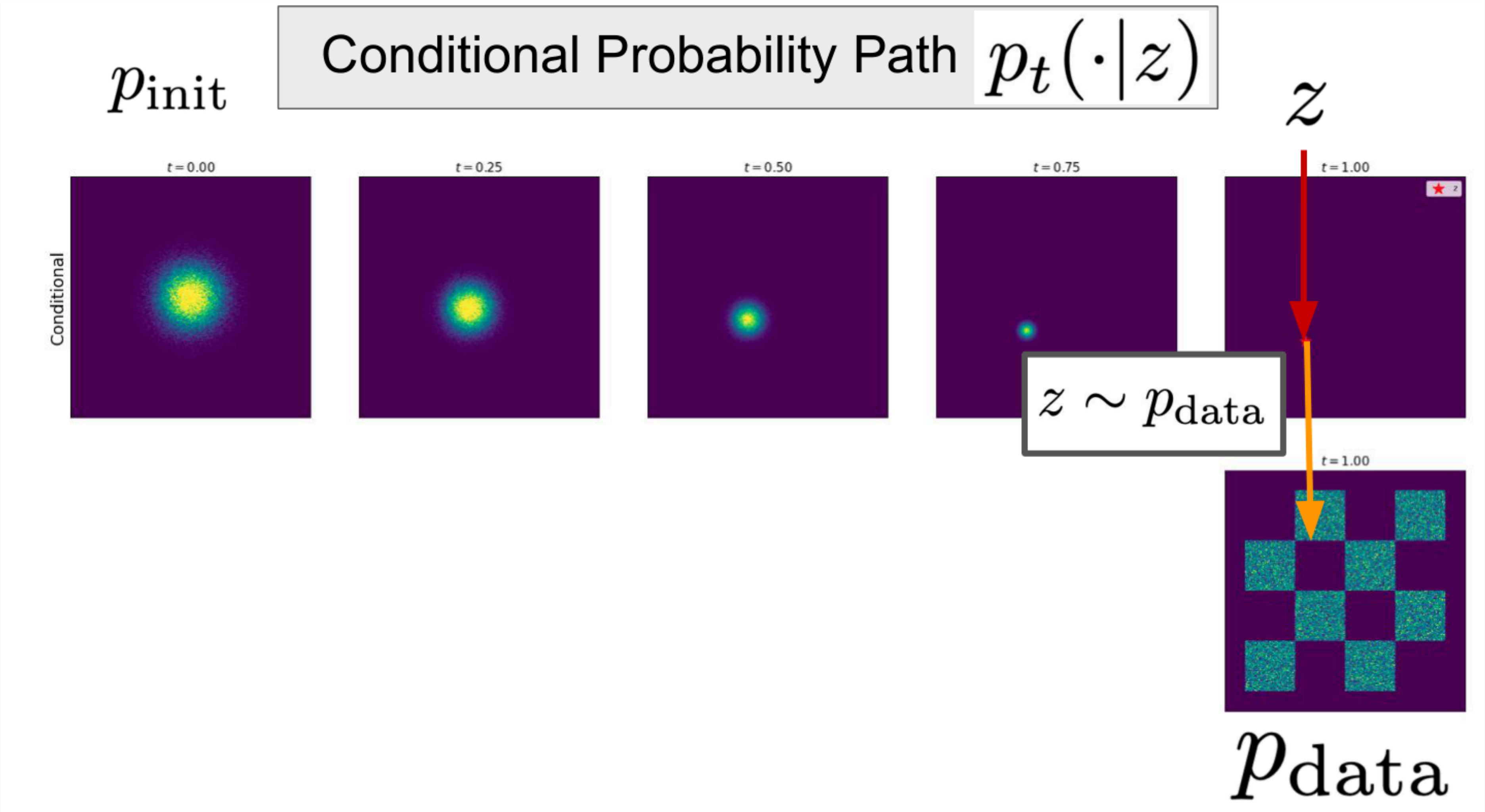

Conditional Probability Path $p_t(\cdot|z)$
p_{init}
z
t = 0.00
t = 0.25
t = 0.50
t = 0.75
t = 1.00
Conditional
t=0
t=1

Normalizing Flows — Limitations (source: course materials)

Diffusion Intuition

Forward Process

- Gradually add Gaussian noise: $q(x_t|x_{t-1})$
- Closed form $q(x_t|x_0) = \mathcal{N}\left(\sqrt{\overline{\alpha}_t}x_0, (1 - \overline{\alpha}_t)I\right)$
- Ends at pure noise


Conditional Probability Path $p_t(\cdot|z)$
p_{init}
z
$t = 0.00$
$t = 0.25$
$t = 0.50$
$t = 0.75$
$t = 1.00$
Conditional
$z \sim p_{\text{data}}$
$t = 1.00$
p_{data}

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Diffusion Intuition — Forward Process (source: course materials)

Reverse Process

- Learn to denoise step by step
- $p_{\theta(x_{t-1}|x_t)}$ parameterized by neural net
- Sampling walks from noise to data

Conditional Prob. Path

	Notation	Key property	Gaussian example
Conditional Probability Path	$p_t(\cdot\|z)$	Interpolates p_{init} and a data point z	$\mathcal{N}(\alpha_t z, \beta_t^2 I_d)$
Conditional Vector Field			

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Diffusion Intuition — Reverse Process (source: course materials)

DDPM Objective

- Predict noise ε added at step t
- Simple MSE on ε with random t
- Equivalent variants: predict x_0 or score

Example - Conditional Vector Field for Gaussian

$$\boxed{u_t^{\text{target}}(x|z) = \left(\dot{\alpha}_t - \frac{\dot{\beta}_t}{\beta_t}\alpha_t\right) z + \frac{\dot{\beta}_t}{\beta_t} x}$$

Proof Sketch:

Step 1: By checking ODE, show that the flow of the vector field is given by

$$\psi_t^{\text{target}}(x_0|z) = \alpha_t z + \beta_t x_0$$

Step 2: If $X_0 = x_0 \sim \mathcal{N}(0, I_d)$ is random, then we know that then:

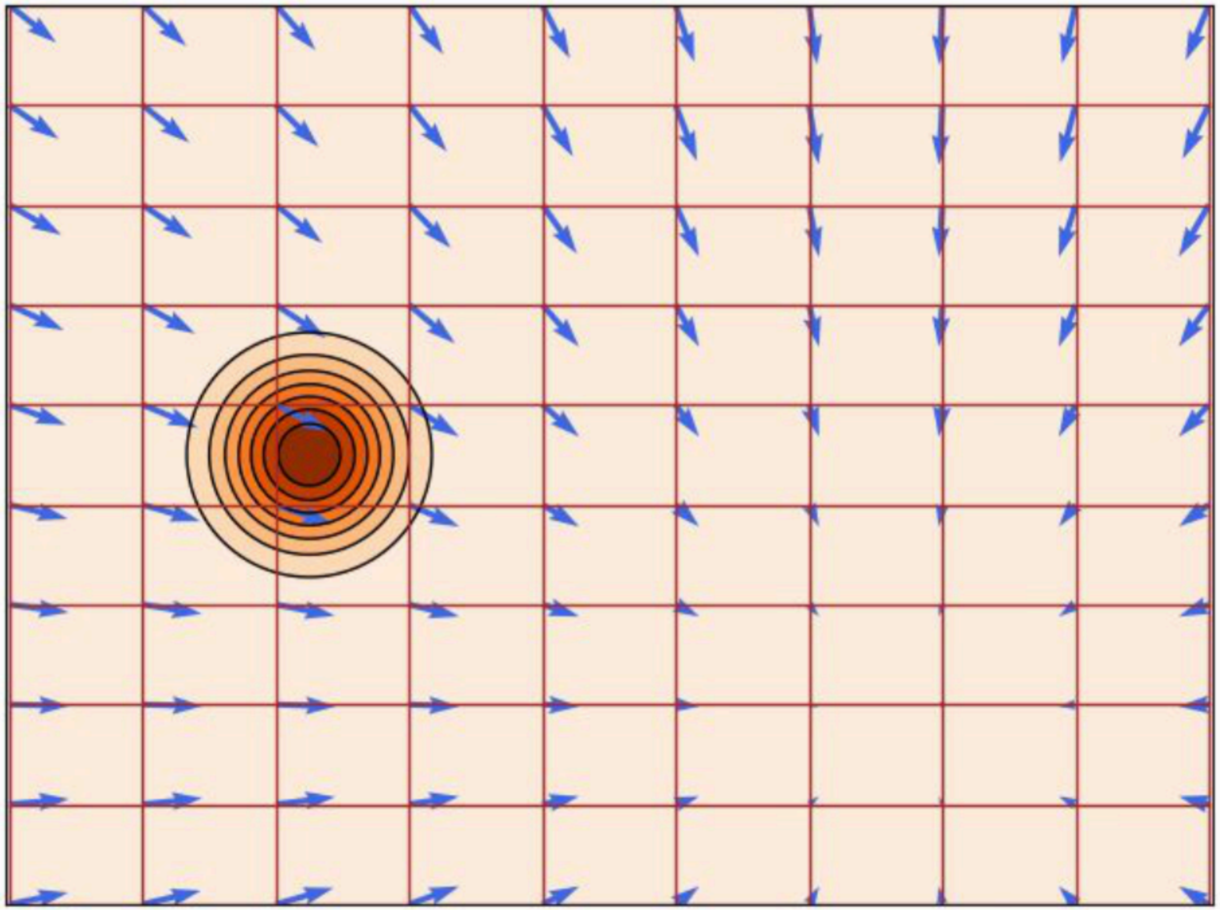
$$X_t = \psi_t(X_0|z) = \alpha_t z + \beta_t X_0 \sim \mathcal{N}(\alpha_t z, \beta_t^2 I_d) = p_t(\cdot|z)$$

Diffusion Intuition — DDPM Objective (source: course materials)

Noise Schedules

- Linear, cosine $\overline{\alpha}_t$ schedules
- Affects training stability and sample quality
- Signal-to-noise ratio view

Simulating ODE with Conditional Vector Field for Conditional Probability Path



NOTE: This is an animated gif and is static in a PDF

Figure credit: Yaron Lipman

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Diffusion Intuition — Noise Schedules (source: course materials)

Score Matching View

Denoising Score Matching

- Learn $s_{\theta(x_t,t)} \approx \nabla_{x_t} \log p(x_t)$
- Tweedie's formula links ε and score
- Unifies diffusion training objectives

Simulating ODE with Marginal Vector Field for Gaussian Probability Path

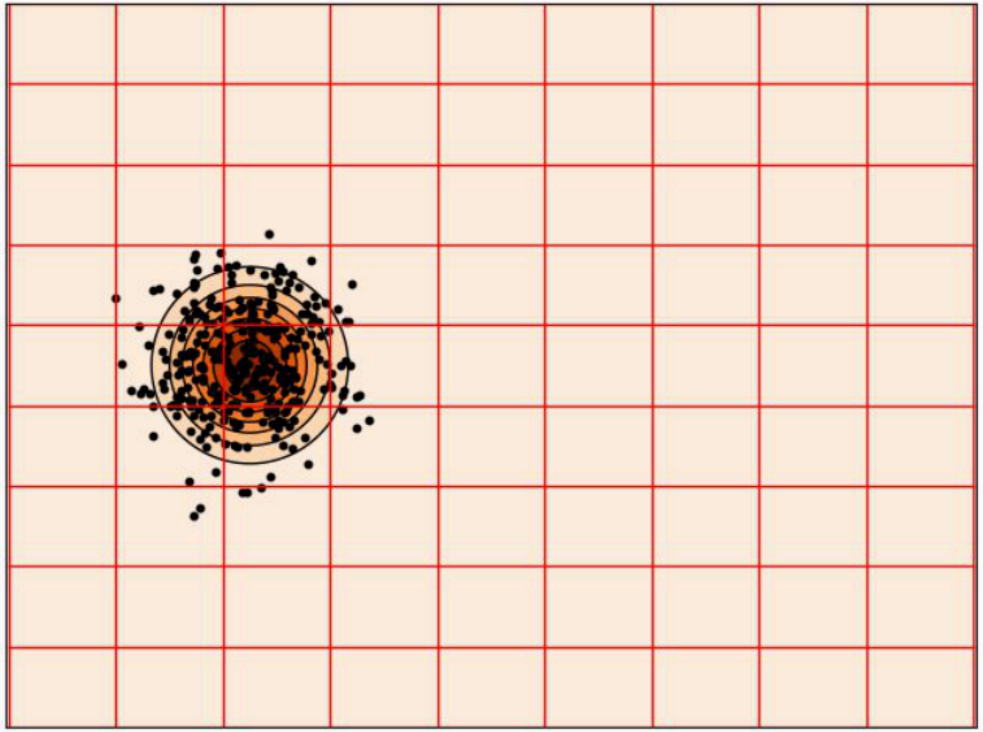

Figure credit: Yaron Lipman

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Score Matching View — Denoising Score Matching (source: course materials)

SDE Formulation Preview

- Forward SDE adds noise continuously
- Reverse SDE uses score function
- Day 8 inference details

Continuity Equation

Randomly initialized ODE

Given: $X_0 \sim p_{\text{init}}, \quad \frac{\mathrm{d}}{\mathrm{d}t} X_t = u_t(X_t)$

Follow probability path:

$$X_t \sim p_t \quad (0 \leq t \leq 1)$$

Marginals are p_t

equivalent

Continuity equation holds

$$\frac{\mathrm{d}}{\mathrm{d}t} p_t(x) = -\mathrm{div}(p_t u_t)(x)$$

PDE holds

Score Matching View — SDE Formulation Preview (source: course materials)

Classifier-Free Guidance

- Train conditional model with dropped labels
- Guidance scale trades diversity vs fidelity
- Standard in text-to-image systems

Lecture 03

Score Matching and Guidance

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Score Matching View — Classifier-Free Guidance (source: course materials)

Latent Diffusion

- Diffuse in VAE latent space (Stable Diffusion)
- Lower dimension $\longrightarrow$ cheaper training
- Text encoder provides conditioning

Algorithm 3 Flow Matching Training Procedure (General)

Require: A dataset of samples $z \sim p_{\text{data}}$, neural network u_t^θ

1: **for** each mini-batch of data **do**

2: Sample a data example z from the dataset.

3: Sample a random time $t \sim \text{Unif}_{[0,1]}$.

4: Sample $x \sim p_t(\cdot|z)$

5: Compute loss

$$\mathcal{L}(\theta) = \|u_t^\theta(x) - u_t^{\text{target}}(x|z)\|^2$$

6: Update the model parameters θ via gradient descent on $\mathcal{L}(\theta)$

7: **end for**

We can learn the marginal vector field by approximating the cond. VF for many different data points z.

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Score Matching View — Latent Diffusion (source: course materials)

Training Practice

Network Architecture

- U-Net with time embedding t
- Attention at lower resolutions
- GroupNorm + SiLU activations

Compute & Data

- Large-scale image-text pairs
- Mixed precision and EMA weights
- Checkpointing long runs

Monitoring

- Loss per noise level
- Periodic sample grids
- FID on small val set

Summary

- Day 7: **Training Flow & Diffusion Models**
- Continuous transforms and denoising objectives
- Questions welcome — practical follows

Questions?

Practical session follows — see course site.